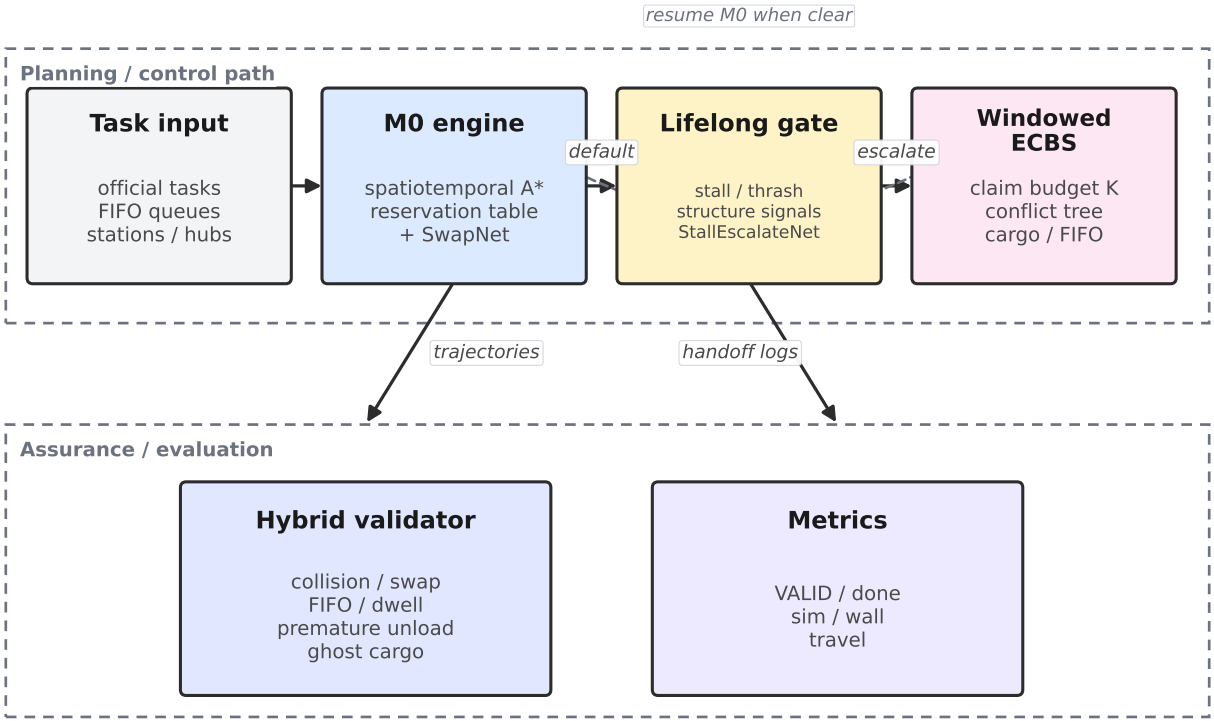Hierarchical Lifelong MAPD Stack